

Homework 2

ASEN 5335 Aerospace Environment

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Problem 1: Plotting the Ionosphere

(i.)

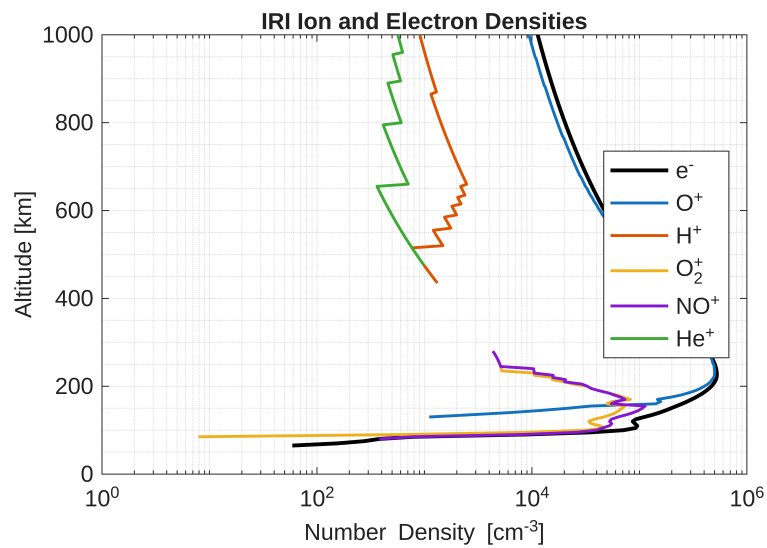


Figure 1: IRI Output

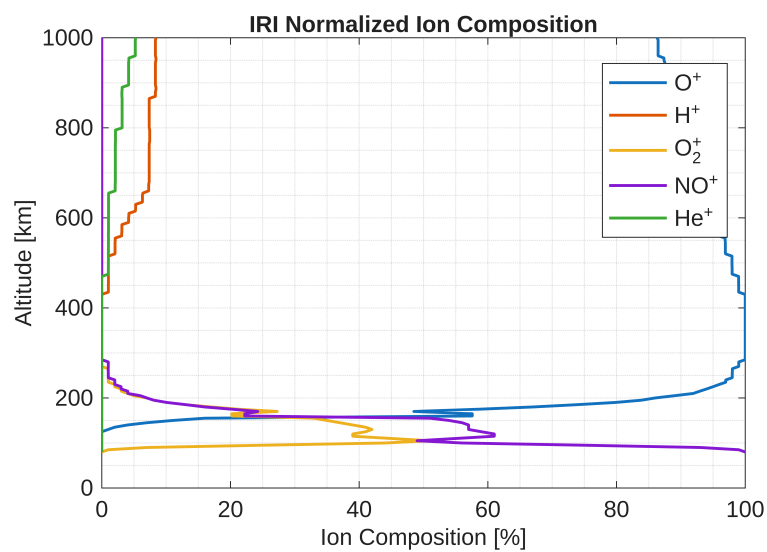


Figure 2: IRI Output Normalized

(ii.)

Below 500 km, we can observe that NO^+ dominates between 0-160 km. O_2^+ contributes more than 20% at between 90 and 175 km.

(iii.)

By convention 1 TECU is 10^{16} el/m². Integrating the total electrons from the IRI output, we find the TEC of the entire IRI output to be 12.47 TECU. When the peak electron density is 525696 m^{-3} , we can integrate the topside and bottomside separately to find they are 3.55 TECU and 8.92 TECU respectively. Normalizing against the total TEC found, we find they are 28.49% and 71.51% respectively.

Problem 2: Peak Ionization

Problem 3: Spacecraft Charging in LEO

Problem 4: Spacecraft Charging in GEO

Problem 5: Radio Wave Propagation

Problem 6: Radio Blackout

Problem 7: GPS Delays

Problem 8: D Region Signal Attenuation

Problem 9: ICON Mission